How do you make indoor solar cells, and what makes them unique?

We use organic solar cells at Epishine. 'Organic' means that the semiconductor, or active layer, used for these solar cells is based on hydrocarbons. The technology we work with is polymer solar cells where we in the current version use a fine blend of long polymer chains and small molecules called fullerenes, in what can be compared to a very thin layer of spaghetti and meatballs. Our solar cells are like a sandwich, with the bread being protective plastic films with a number of thin layers in between, where the spaghetti and meatball layer is in the middle. Between the plastic film and the active layer, there is also an electrode that transports photo current to the contacts and other thin layers that help to direct the photo current within the solar cell.

We make these layers using printing and coating, mainly screen printing and slot-die coating, in a roll-to-roll process. What is special is our lamination method. We put electrode and polymer layers on two separate substrates, then join them with heat and pressure. This gives us good quality and works well in low light. Compared to older indoor solar cells, like in calculators, ours perform better and are more robust.

Can you tell us what organic solar cells are?

Organic solar cells use special materials from nature to turn sunlight into electricity. We are using materials that are based on carbon. We use PET substrates; which are about 200 microns thick, and the solar cell itself is less than a micrometre. The cell material is mainly conjugated organic molecules, and the main component is the PET substrate. There is also a small silver print on the side for conducting current from the cell to the contacts.

How does your technology turn indoor light into electricity?

It is about using the photovoltaic effect, like regular solar cells, but for indoor light from LEDs or fluorescent sources. Indoor light is much weaker than outdoor light, often 100 to 1000 times less intense. Our cells focus on absorbing visible light, unlike silicon cells that also absorb near-infrared for more power. A key for indoor cells is high parallel resistance or rather to prevent short circuits, especially important in low light. This prevents performance drops. So, similar to standard solar cells, but with a focus on eliminating also minor short circuits for indoor effectiveness.

What steps have you taken to increase their efficiency in low light?

Our main goal has been to launch our product, focusing on organic solar cells. Currently, the active layer uses a conjugated polymer mixed with fullerenes, which are like small carbon spheres with a functional tail, allowing for decent performance. However, our R&D team is working to replace these fullerenes with different small molecules, which could potentially double our solar cells' power conversion efficiency. This involves tweaking some components in the active layer, using what is known as non-fullerene acceptors in academic circles. Our aim is to improve power conversion efficiency through these changes.

How are organic photovoltaics different from regular solar cells and how do they help IoT applications?

Our innovation benefits IoT applications because it offers high, consistent performance, especially important for indoor IoT products where light varies a lot. You can trust the solar cell's minimum performance and only need to consider light intensity changes based on placement. Additionally, our solar cells are lightweight, only 0.2 millimetres thick, have low environmental impact, and are easy to integrate.

Could you provide data or results from real-world indoor performance testing?

The indoor solar cell measurements typically yield around 20 to 22 microwatts per square centimetre at 500 lux (LED illumination) in an office setting. We collaborate with companies like ELCs for products like temperature and humidity sensors. We have data on light intensity expectations for various environ-